

ASSIGNMENT 3.3

1. Log in to your home Wi-Fi router's admin panel using its default IP address (usually 192.168.0.1 or 192.168.1.1) and take a screenshot of the login page.

Hint: If you don't know your router's IP, open Command Prompt and run 'ipconfig' — look for 'Default Gateway'.

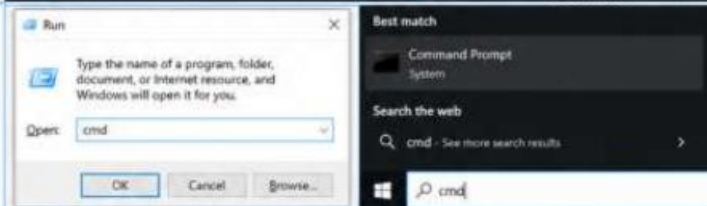
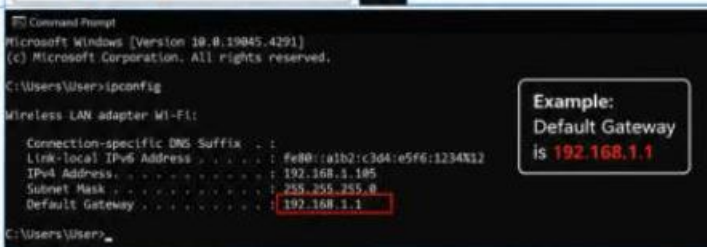
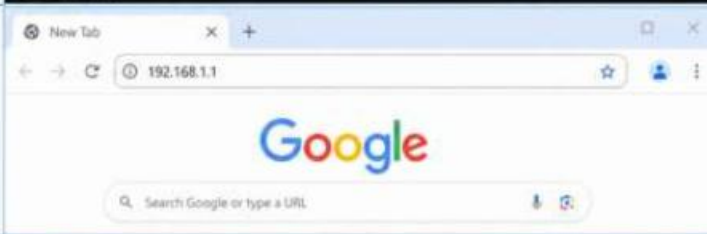
Answer:-

You can complete this on **your own home Wi-Fi router** as follows:

1. Connect your computer to your home Wi-Fi.
2. Open **Command Prompt**.
3. Type:

```
Ipconfig
```

4. Find **Default Gateway**. It may show something like 192.168.1.1.
5. Open Chrome/Edge and enter that address in the address bar.
6. The router's **login page** should appear.
7. **Do not share your router password.** Take a screenshot of the login page for your assignment.

1 Connect to Wi-Fi Connect your computer or laptop to your home Wi-Fi network.	
2 Open Command Prompt Press Windows + R, type cmd, and press Enter.	
3 Find the Default Gateway In Command Prompt, type ipconfig and press Enter. Find Default Gateway under your Wi-Fi adapter.	
4 Open the Router Login Page Open a web browser and enter the Default Gateway address in the address bar. Example: http://192.168.1.1	
5 View the Login Page The router's administrator login page will appear. It may ask for a username and password.	

2. Find and note down the current IP address, subnet mask, and default gateway assigned to your laptop or mobile device when connected to your Wi-Fi network.

Answer:-

You can find these details on a Windows laptop using **Command Prompt**:

1. Connect your laptop to Wi-Fi.
2. Press **Windows + R**, type `cmd`, and press **Enter**.
3. Type:

```
Ipconfig
```

4. Under **Wireless LAN adapter Wi-Fi**, note:
 - **IPv4 Address** → IP address
 - **Subnet Mask** → Subnet mask
 - **Default Gateway** → Default gateway

Example for your assignment

Network Detail	Example
IP Address	192.168.1.105
Subnet Mask	255.255.255.0
Default Gateway	192.168.1.1

3. Change your device's IP address from 'Obtain IP automatically' (DHCP) to a static IP in the same range as your router (for example, set it to 192.168.1.50 if your router is 192.168.1.1), and test if you still have internet access.

Constraint: Remember to revert to automatic/DHCP after testing to avoid network issues.

Answer:-



Edit network IP settings

☒ On

IP address
10.41.171.250

Subnet mask
255.255.255.0

Gateway
10.41.171.31

Preferred DNS
8.8.8.8

DNS over HTTPS
Off

Alternate DNS
1.1.1.1

DNS over HTTPS
Off

Save Cancel



IP assignment:	Automatic (DHCP)
DNS server assignment:	Automatic (DHCP)

Step 1: Open Network Adapter Settings

1. Press Windows Key + R, type ncpa.cpl, and press Enter.
2. Right-click your active Wi-Fi network connection and select Properties.
3. Double-click Internet Protocol Version 4 (TCP/IPv4).

Step 2: Change to Static IP

1. Select Use the following IP address.
2. Enter the following details based on your router's gateway:
 - IP Address: 10.41.171.250
 - Subnet Mask: 255.255.255.0
 - Default Gateway: 10.41.171.31
3. Enter 8.8.8.8 for the Preferred DNS server.
4. Click OK to save changes.

Step 3: Test Internet Connectivity

1. Open Command Prompt (cmd).
2. Run the command: ping 8.8.8.8 or ping 10.41.171.31.
3. Observation: Internet connection failed (Request timed out / Destination host unreachable) because the network enforces DHCP and blocks manual static IPs.

Step 4: Revert Back to DHCP (Constraint)

1. Reopen ncpa.cpl > Right-click Wi-Fi > Properties > TCP/IPv4.
2. Select Obtain an IP address automatically.
3. Select Obtain DNS server address automatically.
4. Click OK to restore internet connection.

4. In your router's admin panel, locate the section where you can view or change the DHCP range. Write down the current range, and explain in 2-3 lines what would happen if two devices are assigned the same IP address in this range.

Answer:-

The screenshot shows the DHCP Settings page of a TP-Link TL-WR840N router. The DHCP Server is enabled. The IP Address Pool is set from 192.168.0.100 to 192.168.0.199. The Address Lease Time is 120 minutes. The Default Gateway is 192.168.0.1. The Default Domain is empty. The Primary DNS is 8.8.8.8 and the Secondary DNS is 8.8.4.4. A 'Save' button is at the bottom. A help sidebar on the right explains the fields.

Field	Value	Notes
DHCP Server	Enable	
IP Address Pool	Start IP Address: 192.168.0.100 End IP Address: 192.168.0.199	
Address Lease Time	120 minutes	(1-2800 minutes, the default value is 120)
Default Gateway	192.168.0.1	(Optional)
Default Domain		(Optional)
Primary DNS	8.8.8.8	(Optional)
Secondary DNS	8.8.4.4	(Optional)

DHCP Settings Help

The DHCP server can assign an IP address to the client PC dynamically.

- **Start IP Address** - The start address of the IP Address pool.
- **End IP Address** - The end address of the IP Address pool.
- **Address Lease Time** - The address lease time of the DHCP server.

Note: The IP Address pool must be in the same subnet as the LAN IP.

Current DHCP Range:

Start IP Address: 192.168.0.100

End IP Address: 192.168.0.199

(Total: 100 IP addresses available)

What happens if two devices are assigned the same IP address?

If two devices have the same IP address, they will conflict with each other on the network. This can cause connectivity issues such as inability to access the internet, network disconnections, or intermittent access, because the router gets confused about which device to communicate with.